

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Substitution: energy transferred = 50×150 (1) = 7500 [J] (1)	1	1		2	2	2
		(ii)		Change in temperature = 15 [°C]		1		1		1
		(iii)		Substitution: $\frac{7500 \text{ ecf}}{[1 \times] 15 \text{ ecf}}$ (1) $c = 500$ [J/kg °C] (1)	1	1		2	2	2
		(iv)	I	Difference = 70 [J/kg °C]		1		1	1	1
			II	Substitution: $\frac{70 \text{ ecf}}{385} \times 100$ (1) % difference = 18.2 or 18 or 18.18 (1) Award 1 mark only for 18.1 or 0.18	1	1		2	2	2
	(b)			Ticks in boxes 3 and 4: Wrap block in insulating material (1) push heater fully into block (1) Deduct 1 mark for any additional tick			2	2		2
				Question 4 total	3	5	2	10	7	10